

Lin Wang 王林

Postdoctoral Researcher - AI for Molecular Discovery

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Lin Wang develops AI methods for molecular representation, protein modeling, and drug discovery. Current research focuses on molecular foundation models, conformational-space and pharmacophore-aware representation learning, phenotype-based screening, and structure generation. Recent work includes Ouroboros, a property-guided molecular generation framework published in *Advanced Science*, and earlier systems for protein interaction mining, synthetic accessibility prediction, and virtual screening.

Research Focus

- Molecular representation and generation foundation models for medical chemistry
- Conformational-space and pharmacophore-aware molecular representations
- Protein-ligand and protein-protein interaction modeling
- Phenotype-based and poly-pharmacology drug design

Education and Experience

Current	Postdoctoral Researcher Institute of Systems Medicine, Chinese Academy of Medical Sciences, Suzhou, China. Molecular and protein foundation models, phenotype-based learning, and translational drug discovery.
2020-2024	Ph.D., ShanghaiTech University Structural biology, molecular dynamics, protein interactions, and physically informed molecular representation.
2019	B.S. in Life Science, Northeast Agricultural University Biological foundations and early computational drug-discovery workflows.

Selected Research Projects

2026	Ouroboros Property-guided molecular generation in a learned encoding space. A shared representation supports molecular graphs, property prediction, and SMILES generation. Advanced Science paper - GitHub repository
2024	GeminiMol Conformational-space profiling for general molecular representation. Hybrid contrastive learning integrates conformational and pharmacophore similarity. Using GeminiMol, we identified an inhibitor of GluN1/GluN3A ($IC_{50} = 0.98 \mu M$); the work received first prize in the 2023 Shanghai International Computational Biology Innovation Competition. Advanced Science paper - GitHub repository - Application: Discovery of novel GluN1/GluN3A NMDA receptor inhibitors using a deep learning-based method
2022	PPI-Miner Structure and sequence motif co-driven method for protein-protein interaction mining and modeling. Applied to molecular glue substrate discovery, leading to a curated library of potential CRBN protein substrates. J. Chem. Inf. Model. paper - GitHub repository - CRBN substrate library

Publications

= co-first author; * = co-corresponding author

2026

Integrating chemical priors and physical laws to mitigate hallucinations in structure-based drug design

Zhongyu Liu, Yadong Liu*, Yihang Zhou, Zhiyuan Liu, Yuhang Yang, Zhiwen Wang, Tianyi Zang*, **Lin Wang***, Yang Zhang*
Communications Chemistry, 2026 - Google Scholar

Learned Conformational Space and Pharmacophore Into Molecular Foundational Model

Lin Wang, Yifan Wu, Hao Luo, Minglong Liang, Yihang Zhou, Cheng Chen, Chris Liu, Jun Zhang, Yang Zhang
Advanced Science, p. e13556, 2026 - DOI

Molecular glue degraders of HuR suppress BRAF-mutant colorectal cancer

Xiaocui Lu, Xiuyun Wang, Zheng Yang, Xusheng Wang, **Lin Wang**, Chunhui Xu, I Lo, Chenlu Geng, Yisheng Pu, Keyu Zhang, others
Nature, p. 1-10, 2026 - Google Scholar

ProteinConformers: Benchmark Dataset for Simulating Protein Conformational Landscape Diversity and Plausibility

Yihang Zhou[#], Chen Wei[#], Minghao Sun, Jin Song, Yang Li*, **Lin Wang***, Yang Zhang*
Advances in Neural Information Processing Systems, vol. 38, 2026 - Google Scholar

Small Molecule Conformation Prediction

Lin Wang, Fang Bai
Artificial Intelligence for Drug Design, p. 527-546, book chapter, 2026 - Google Scholar

2025

DeepPSA: A Geometric Deep Learning Model for PROTAC Synthetic Accessibility Prediction

Ran Zhang, Shihang Wang, **Lin Wang**, Siyuan Tian, Yilin Tang, Fang Bai
Journal of Chemical Information and Modeling, vol. 65, p. 6861-6873, 2025 - Google Scholar

Discovery of novel GluN1/GluN3A NMDA receptor inhibitors using a deep learning-based method

Shi-hang Wang, Yue Zeng, Hao Yang, Si-yuan Tian, Yong-qi Zhou, **Lin Wang**, Xue-qin Chen, Hai-ying Wang, Zhao-bing Gao*, Fang Bai*
Acta Pharmacologica Sinica, vol. 46, p. 3108-3115, 2025 - Google Scholar

Fitness costs of mobilised colistin resistance gene 3 (mcr-3): systematic review, epidemiological study, and functional analysis

Lujie Liang[#], Yaxin Li[#], **Lin Wang[#]**, Wenli Wang, Yihao Zhang, Hui Zhao, Yaxuan Wang, Lingxuan Lyu, Jiachen Li, Dianrong Zhou, Zhe Hu, Lizhen Luo, Guanxiu Wang, Jia Wan, Lin Xu, Meisong Li, Min Dai, Meiting Yang, Shun Xiong, Lan-Lan Zhong, Fang Bai, Siyuan Feng, Guo-Bao Tian
EBioMedicine, vol. 120, p. 105923, 2025 - DOI

MAI-TargetFisher: A proteome-wide drug target prediction method synergetically enhanced by artificial intelligence and physical modeling

Shi-wei Li, Peng-xuan Ren, **Lin Wang**, Qi-lei Han, Feng-lei Li, Hong-lin Li, Fang Bai
Acta Pharmacologica Sinica, vol. 46, p. 1462-1475, 2025 - Google Scholar

NADPH-Independent Fluorescent Probe for Live-Cell Imaging of Heme Oxygenase-1

Liang Li, Xuanyi Lu, Qiyuan He, Chao Shu, Edward RH Walter*, **Lin Wang***, Nicholas J Long*, Lijun Jiang*
ACS sensors, vol. 10, p. 499-506, 2025 - Google Scholar

PhenoModel: A multimodal phenotypic drug design foundation model for discovering novel potential inhibitors of multiple cancer cells

Shihang Wang[#], Qilei Han[#], Weichen Qin[#], **Lin Wang[#]**, Junhong Yuan, Fengyu Cai, Yiqun Zhao, Pengxuan Ren, Yunze Zhang, Yilin Tang, others
Acta Pharmaceutica Sinica B, 2025 - DOI

Recent advances in molecular representation methods and their applications in scaffold hopping

Shihang Wang, Ran Zhang, Xiangcheng Li, Fengyu Cai, Xinyue Ma, Yilin Tang, Chao Xu, **Lin Wang**, Pengxuan Ren, Lu Liu, others
Npj Drug Discovery, vol. 2, p. 14, 2025 - Google Scholar

2024

Conformational space profiling enhances generic molecular representation for AI-powered ligand-based drug discovery

Lin Wang, Shihang Wang, Hao Yang, Shiwei Li, Xinyu Wang, Yongqi Zhou, Siyuan Tian, Lu Liu, Fang Bai

Advanced Science, vol. 11, p. 2403998, 2024 - DOI

Salvianolic acid B attenuates liver fibrosis by targeting Ecm1 and inhibiting hepatocyte ferroptosis

Yadong Fu, Xiaoxi Zhou, Lin Wang, Weiguo Fan, Siqi Gao, Danyan Zhang, Zhiyang Ling, Yaguang Zhang, Liyan Ma, Fang Bai, others

Redox biology, vol. 69, p. 103029, 2024 - Google Scholar

2023

A new variant of the colistin resistance gene MCR-1 with co-resistance to β -lactam antibiotics reveals a potential novel antimicrobial peptide

Lujie Liang[#], Lan-Lan Zhong[#], Lin Wang[#], Dianrong Zhou, Yaxin Li, Jiachen Li, Yong Chen, Wanfei Liang, Wenjing Wei, Chenchen Zhang, others

PLoS Biology, vol. 21, p. e3002433, 2023 - Google Scholar

DeepSA: a deep-learning driven predictor of compound synthesis accessibility

Shihang Wang[#], Lin Wang[#], Fenglei Li, Fang Bai

Journal of Cheminformatics, vol. 15, p. 103, 2023 - DOI

Discovery, evaluation and mechanism study of WDR5-targeted small molecular inhibitors for neuroblastoma

Qi-lei Han, Xiang-lei Zhang, Peng-xuan Ren, Liang-he Mei, Wei-hong Lin, Lin Wang, Yu Cao, Kai Li^{*}, Fang Bai^{*}

Acta Pharmacologica Sinica, vol. 44, p. 877-887, 2023 - Google Scholar

Discovery, structural optimization, and anti-tumor bioactivity evaluations of betulinic acid derivatives as a new type of ROR γ antagonists

Lianghe Mei, Lansong Xu, Sanan Wu, Yafang Wang, Chao Xu, Lin Wang, Xingyu Zhang^{*}, Chengcheng Yu, Hualiang Jiang, Xianglei Zhang^{*}, others

European Journal of Medicinal Chemistry, vol. 257, p. 115472, 2023 - Google Scholar

Propofol exerts anti-anhedonia effects via inhibiting the dopamine transporter

Xiao-Na Zhu, Jie Li, Gao-Lin Qiu, Lin Wang, Chen Lu, Yi-Ge Guo, Ke-Xin Yang, Fang Cai, Tao Xu^{*}, Ti-Fei Yuan^{*}, others

Neuron, vol. 111, p. 1626-1636, 2023 - Google Scholar

Synthesis of maleimide-braced peptide macrocycles and their potential anti-SARS-CoV-2 mechanisms

Jian Li, Jina Sun, Xianglei Zhang, Ruxue Zhang, Qian Wang, Lin Wang, Leike Zhang, Xiong Xie, Chunpu Li, Yu Zhou, others

Chemical Communications, vol. 59, p. 868-871, 2023 - Google Scholar

2022

A multi-targeting drug design strategy for identifying potent anti-SARS-CoV-2 inhibitors

Peng-xuan Ren, Wei-juan Shang, Wan-chao Yin, Huan Ge, Lin Wang, Xiang-lei Zhang^{*}, Bing-qian Li, Hong-lin Li, Ye-chun Xu, Eric H Xu, others

Acta Pharmacologica Sinica, vol. 43, p. 483-493, 2022 - Google Scholar

Crystal structures of Wolbachia CidA and CidB reveal determinants of bacteria-induced cytoplasmic incompatibility and rescue

Haofeng Wang, Yunjie Xiao, Xia Chen, Mengwen Zhang, Guangxin Sun, Feng Wang, Lin Wang, Hanxiao Zhang, Xiaoyu Zhang, Xin Yang, others

Nature Communications, vol. 13, p. 1608, 2022 - Google Scholar

Discovery of potential small molecular SARS-CoV-2 entry blockers targeting the spike protein

Lin Wang[#], Yan Wu[#], Sheng Yao[#], Huan Ge[#], Ya Zhu, Kun Chen, Wen-zhang Chen, Yi Zhang, Wei Zhu^{*}, Hong-yang Wang, others

Acta Pharmacologica Sinica, vol. 43, p. 788-796, 2022 - Google Scholar

PiLSL: pairwise interaction learning-based graph neural network for synthetic lethality prediction in human cancers

Xin Liu, Jiale Yu, Siyu Tao, Beiyuan Yang, Shike Wang, Lin Wang, Fang Bai, Jie Zheng

Bioinformatics, vol. 38, p. ii106-ii112, 2022 - Google Scholar

PPI-Miner: a structure and sequence motif co-driven protein-protein interaction mining and modeling computational method

Lin Wang, Feng-lei Li, Xin-yue Ma, Yong Cang, Fang Bai

Journal of Chemical Information and Modeling, vol. 62, p. 6160-6171, 2022 - DOI

Rapamycin targets STAT3 and impacts c-Myc to suppress tumor growth

Le Sun, Yu Yan, Heng Lv, Jianlong Li, Zhiyuan Wang, Kun Wang, **Lin Wang**, Yunxia Li, Hong Jiang, Yaoyang Zhang

Cell chemical biology, vol. 29, p. 373-385, 2022 - Google Scholar

Recent advances in predicting protein-protein interactions with the aid of artificial intelligence algorithms

Shiwei Li, Sanan Wu, **Lin Wang**, Fenglei Li, Hualiang Jiang, Fang Bai

Current Opinion in Structural Biology, vol. 73, p. 102344, 2022 - Google Scholar

Target-based virtual screening and LC/MS-guided isolation procedure for identifying phloroglucinol-terpenoid inhibitors of SARS-CoV-2

Bo Hou, Yu-Min Zhang, Han-Yi Liao, Li-Feng Fu, De-Dong Li, Xin Zhao, Jian-Xun Qi, Wei Yang, Geng-Fu Xiao, Lian Yang, others

Journal of natural products, vol. 85, p. 327-336, 2022 - Google Scholar

2021

CKB inhibits epithelial-mesenchymal transition and prostate cancer progression by sequestering and inhibiting AKT activation

Zheng Wang, Mohit Hulsurkar, Lijuan Zhuo, Jinbang Xu, Han Yang, Samira Naderinezhad, **Lin Wang**, Guoliang Zhang, Nanping Ai, Linna Li, others

Neoplasia, vol. 23, p. 1147-1165, 2021 - Google Scholar

Identification of potential binding sites of sialic acids on the RBD domain of SARS-CoV-2 spike protein

Bingqian Li, **Lin Wang**, Huan Ge, Xianglei Zhang, Penxuan Ren, Yu Guo, Wuyan Chen, Jie Li, Wei Zhu, Wenzhang Chen, others

Frontiers in Chemistry, vol. 9, p. 659764, 2021 - Google Scholar

Probing the allosteric inhibition mechanism of a spike protein using molecular dynamics simulations and active compound identifications

Qian Wang[#], **Lin Wang**[#], Yumin Zhang[#], XiangLei Zhang, Leike Zhang, Weijuan Shang, Fang Bai^{*}

Journal of Medicinal Chemistry, vol. 65, p. 2827-2835, 2021 - Google Scholar

2020

Antimicrobial resistance in clinical Ureaplasma spp. and Mycoplasma hominis and structural mechanisms underlying quinolone resistance

Ting Yang, Lianlian Pan, Ningning Wu, **Lin Wang**, Zhen Liu, Yingying Kong, Zhi Ruan, Xinyou Xie, Jun Zhang

Antimicrobial Agents and Chemotherapy, vol. 64, p. 10-1128, 2020 - Google Scholar

Structure of Mpro from SARS-CoV-2 and discovery of its inhibitors

Zhenming Jin, Xiaoyu Du, Yechun Xu, Yongqiang Deng, Meiqin Liu, Yao Zhao, Bing Zhang, Xiaofeng Li, Leike Zhang, Chao Peng, others

Nature, vol. 582, p. 289-293, 2020 - DOI

The structural study of mutation-induced inactivation of human muscarinic receptor M4

Jingjing Wang, Meng Wu, Lijie Wu, Yueming Xu, Fei Li, Yiran Wu, Petr Popov, **Lin Wang**, Fang Bai, Suwen Zhao, others

IUCrJ, vol. 7, p. 294-305, 2020 - Google Scholar

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